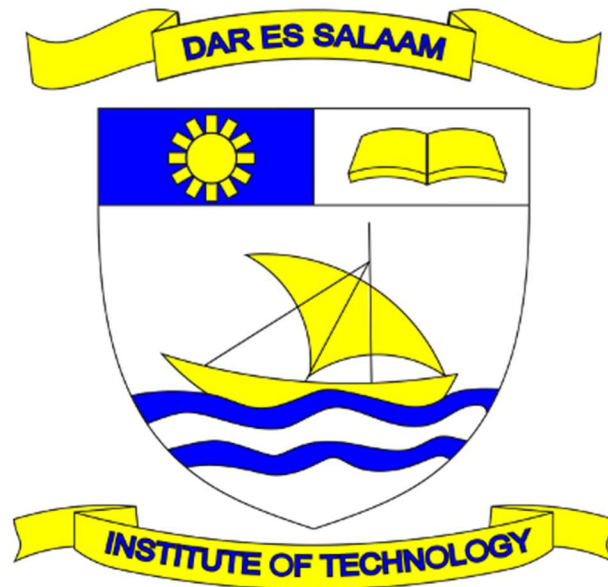


# DAR ES SALAAM INSTITUTE OF TECHNOLOGY



**DEPARTMENT OF COMPUTER STUDIES  
NTA LEVEL 8**

## **PROJECT REPORT**

### **DEVELOPMENT OF A CLOUD-BASED HOTSPOT MANAGEMENT AND BILLING SYSTEM**

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**ACADEMIC YEAR:** 2025/2026

## DECLARATION

I, SHABILU ALLY HAMISI, declare that this project proposal report, titled “Development of a Cloud-Based Hotspot Management and Billing System” is my own original work and research. This report is submitted in partial of a Bachelor in Computer Engineering and has not been copied from anywhere else or presented anywhere. All sources of information used in this study have been acknowledged and cited appropriately in the reference section.

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## **CHAPTER TWO: INTRODUCTION**

### **1.1 Background of the Study**

Access to affordable and reliable Internet connectivity has become essential for individuals and small businesses operating in local marketplaces. In busy commercial areas such as Karume Market in Dar es Salaam, traders, customers, and service providers rely on the Internet for communication, mobile money transactions, online marketing, and access to digital services. Because mobile data bundles are relatively expensive for continuous use, small-scale Wi-Fi hotspot providers have emerged to offer shared Internet access at lower prices.

These providers install wireless routers that allow multiple users within the market to connect using smartphones and laptops. While this offers an affordable alternative to mobile data, the way these hotspot services are managed remains largely manual and decentralized. Each router is configured and operated on its own, billing is handled through cash or hand-written vouchers, and access is controlled using shared passwords. As the number of users and routers grows, these manual practices become increasingly difficult to sustain.

Advances in cloud computing now make it possible to manage distributed network devices from a single centralized platform accessible over the Internet. A cloud-based hotspot management and billing system can therefore centralize user authentication, billing, monitoring, and administration, and when integrated with the mobile money services widely used in Tanzania, can allow users to purchase Internet access conveniently from their phones. This study develops such a system, using hotspot operations in Karume Market as the case study.

### **1.2 Problem Statement**

Wi-Fi hotspot providers operating in marketplace environments such as Karume Market deliver affordable Internet to traders and customers, but they manage their networks using manual, decentralized methods that create serious operational problems.

Each router is configured and operated independently, with no centralized platform to control users, enforce access policies, or oversee performance across multiple devices. Billing is handled manually through cash payments or hand-generated vouchers, which makes the process slow, error-prone, and difficult to reconcile, resulting in inaccurate and delayed billing. Because access depends on shared Wi-Fi passwords with weak authentication, unauthorized users frequently connect without paying, causing revenue leakage for operators.

Operators also have limited ability to monitor bandwidth usage and network performance, so congestion during peak hours goes undetected and unmanaged. When a router fails, there is no automatic alert, and operators often discover outages only when customers complain, leading to prolonged downtime. The lack of centralized management makes it difficult to coordinate multiple routers, while limited remote access forces operators to be physically present to manage their networks. Existing manual approaches also carry high operational and maintenance costs and scale poorly as the number of users and locations increases.

Furthermore, although mobile money services such as M-Pesa and Airtel Money are widely used in Tanzania, current hotspot setups are not integrated with them, reducing payment convenience and increasing the administrative burden on service providers.

Therefore, there is a need for a cloud-based hotspot management and billing system that centralizes user and router management, automates billing through mobile money, monitors bandwidth and router availability in real time, predicts congestion, sends alerts on router failure, supports remote administration, and scales across multiple locations.

### 1.3 Objectives of the Study

#### 1.3.1 Main Objective

To develop a cloud-based hotspot management and billing system.

#### 1.3.2 Specific Objectives

- i. To implement a cloud-based architecture for centralized management of multiple Wi-Fi hotspots, including remote administration and reporting on revenue and network usage.
- ii. To develop a captive portal with an automated billing mechanism that authenticates users and enforces Internet access plans.
- iii. To integrate the system with a mobile money payment platform to enable self-service purchase and renewal of Internet access.
- iv. To implement a router and network monitoring module that reports router availability, connected users, and bandwidth usage, and sends alerts when a router becomes unavailable.
- v. To implement a simple rule-based module that detects and predicts network congestion to support fair bandwidth management.

### 1.4 Significance of the Study

This study addresses the challenges faced by local Wi-Fi hotspot providers in marketplace environments. The proposed system is expected to improve hotspot services in the following ways:

- **Improved network management.** A centralized platform for controlling users, plans, and multiple routers eliminates the need for individual router configuration.
- **Reduced administrative workload.** Automating billing, payment verification, and user management reduces manual effort and administrative errors.
- **Convenient payment for users.** Integration with familiar mobile money services such as M-Pesa and Airtel Money provides a simple payment experience.
- **Reduced revenue leakage.** Proper OTP-based authentication replaces shared passwords, ensuring that only paying users access the network.
- **Improved service reliability.** Real-time router monitoring and automatic failure alerts enable faster response to network problems.

- **Support for scalable deployment.** The centralized cloud architecture enables operators to expand services to multiple locations while maintaining consistent management and monitoring.

### 1.5 Scope of the Study

This study covers the design and development of a cloud-based hotspot management and billing system for Wi-Fi providers operating in marketplace environments such as Karume Market. It includes centralized hotspot management, captive-portal authentication and access control, automated billing with mobile money payment integration, router and bandwidth monitoring with alerts, and a rule-based congestion-detection module. The implementation and evaluation are conducted in a controlled or simulated environment. The study does not cover the deployment of large-scale Internet infrastructure or backbone connectivity provided by Internet Service Providers.

## **CHAPTER TWO: LITERATURE REVIEW**

### **2.1 Introduction**

This chapter reviews existing studies and technologies related to hotspot authentication, Internet billing mechanisms, network monitoring, and cloud-based network management. The purpose is to establish the theoretical and technical foundation for the this system and to identify the research gap it addresses. The chapter discusses the Authentication, Authorization, and Accounting (AAA) framework, reviews previous studies, examines relevant technologies, analyzes the existing manual approach at Karume Market, and presents a comparison between the existing and proposed systems.

### **2.2 The AAA Framework**

Effective hotspot systems rely on a structured access-control model known as Authentication, Authorization, and Accounting (AAA). Authentication verifies the identity of a user attempting to connect to the network. Authorization determines what resources or service plan an authenticated user is allowed to access. Accounting records network usage for monitoring, auditing, and billing purposes. Together these three components provide the foundation for controlled and billable network access. In this project system, OTP phone verification performs authentication, plan-based access performs authorization, and session and usage logging performs accounting.

### **2.3 Review of Previous Studies**

Wiederkehr examined authentication mechanisms used in public wireless networks and analyzed the trade-off between usability and security in hotspot login systems. The study found that most public hotspots rely on browser-based captive-portal authentication because it is simple to deploy and requires no additional software on user devices. It emphasized the role of the AAA framework in managing hotspot access. This project system project adopts the captive-portal approach for the same reasons, while strengthening authentication through one-time passwords rather than shared passwords.

Owoh developed a usage-based billing system that monitors individual data consumption within a university network using MikroTik routers, generating billing reports from collected usage statistics. The study showed that usage-based billing improves fairness in resource allocation. However, the system was designed for a single institutional network and did not address managing multiple routers across different locations. This project extends these billing concepts to a multi-router, multi-location marketplace context and adds automated mobile money payment.

Saliu and colleagues implemented an Internet authentication and billing hotspot system using the MikroTik Router Operating System, demonstrating that captive-portal authentication, bandwidth control, and user monitoring can be enforced at the router level. While effective, such implementations are configured locally on each router, which becomes difficult to coordinate when many routers are deployed across a marketplace. This limitation motivates the centralized cloud architecture used in this project.

On centralized network management, Tanenbaum and Wetherall explain that managing network devices, user access, and policies from a single interface simplifies administration and improves scalability when multiple access points are deployed. Erl and colleagues further show that cloud computing enables administrators to configure devices, monitor performance, and manage users remotely through Internet-accessible platforms. These principles underpin the Proposed system design, in which a central cloud server manages authentication, billing, and monitoring while distributed routers enforce access.

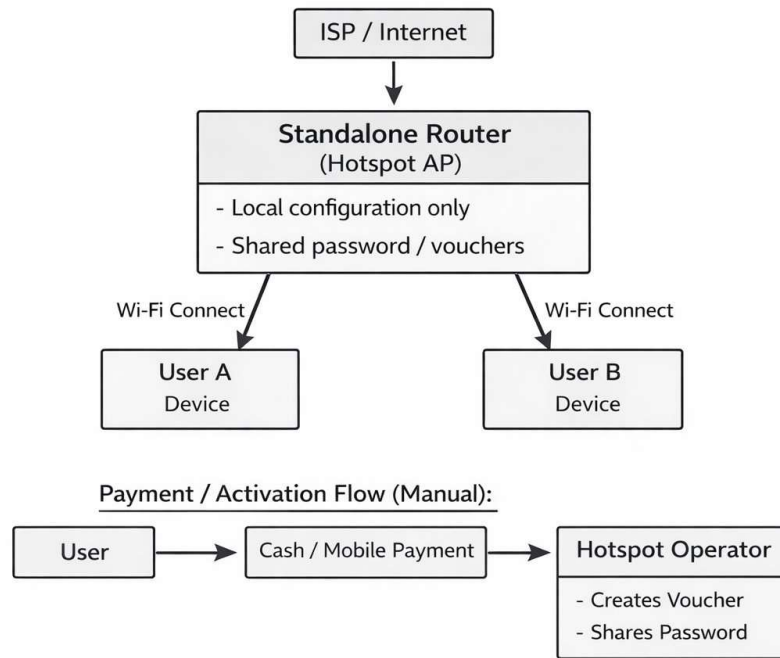
## 2.4 Technologies Used in the Proposed System

- **MikroTik RouterOS.** Used to enforce hotspot access at the router and to create and remove hotspot users, list active users, and read system health
- **FreeRADIUS.** Provides centralized authentication and accounting, with credentials and bandwidth limits provisioned per user from the central database.
- **Captive portal.** Redirects newly connected devices to a billing and login page before granting access, requiring no software installation by the user.
- **Mobile money payment.** Allows users to pay through services such as M-Pesa and Airtel Money and have access activated automatically upon payment confirmation.
- **Cloud hosting.** Provides a central, Internet-accessible server so that operators can manage and monitor routers remotely from any location.

## 2.5 Analysis of the Existing System

In marketplace environments such as Karume Market, Internet connectivity is typically provided by small-scale operators who install wireless routers and control access using shared Wi-Fi passwords or manually generated vouchers. Payment is collected in cash or through informal arrangements. These methods provide weak authentication, allow unauthorized access through password sharing, and make accurate record-keeping difficult.

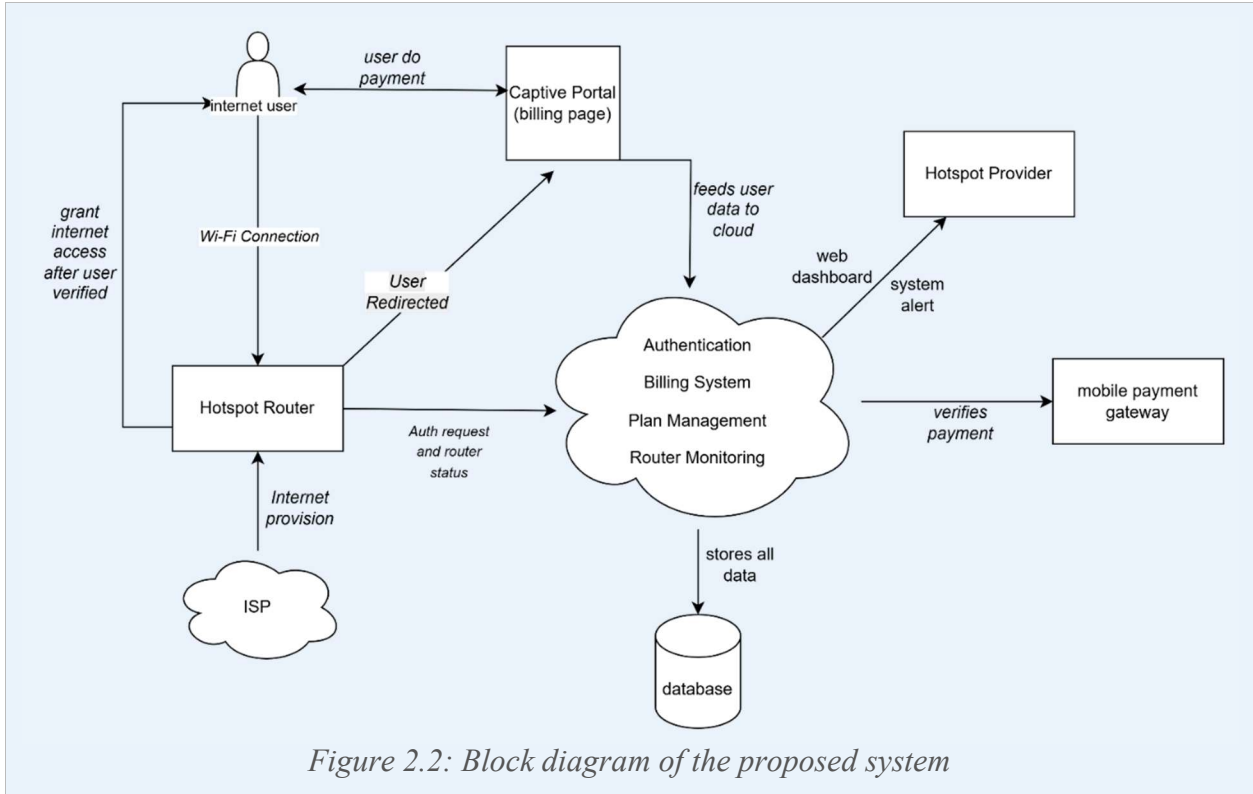
Operators lack real-time visibility into router performance and availability; when a router fails, the problem is often noticed only after users complain. As operators attempt to expand, the absence of centralized remote management makes coordinating multiple routers and enforcing consistent policies increasingly difficult. The key limitations are: lack of centralized management; weak authentication; manual cash or voucher billing; revenue leakage from unauthorized access; no real-time monitoring or failure alerts; limited bandwidth visibility; high operational cost; and poor scalability.



*Figure 2.1: Block diagram of the existing manual hotspot system at Karume Market*

## 2.6 The Proposed System

To overcome these limitations, this study develops proposed system, a cloud-based hotspot management and billing system providing centralized control of multiple hotspot routers. The system adopts the AAA framework for structured authentication, authorization, and accounting, and extends usage-based billing concepts with automated mobile money payment. Its defining feature is a cloud-based centralized architecture that separates management from enforcement: the cloud server handles authentication, billing, usage logging, monitoring, and administration, while the routers enforce access policies.



## 2.7 Comparison: Existing System vs. Proposed system

|                      | <b>existing</b>                    | <b>proposed</b>                           |
|----------------------|------------------------------------|-------------------------------------------|
| Architecture         | Routers operate independently      | Centralized cloud-based management        |
| Authentication       | Shared passwords / manual vouchers | Captive portal with OTP verification      |
| Billing              | Cash or manual vouchers            | Automated billing via mobile money        |
| Payment verification | Manual by operator                 | Automatic via mobile money API            |
| Access control       | Limited                            | Plan-based time and bandwidth limits      |
| Monitoring           | Minimal, per-router                | Centralized monitoring of all routers     |
| Bandwidth visibility | Little or none                     | Usage tracked per user and router         |
| Failure alerts       | None — noticed via complaints      | Automatic alerts when router goes offline |

|                     | <b>existing</b>     | <b>proposed</b>                       |
|---------------------|---------------------|---------------------------------------|
| Congestion handling | None                | Rule-based congestion detection       |
| Remote management   | Must be on-site     | Full remote management over Internet  |
| Reporting           | Manual, if any      | Automated revenue and usage reports   |
| Scalability         | Difficult to expand | Easily scalable to multiple locations |

## 2.8 Knowledge Gap

The reviewed studies provide valuable insights into hotspot authentication, usage-based billing, and centralized network management, but they generally address these components separately and within single-location environments. Few integrate hotspot management, billing automation, monitoring, and payment into one platform, and limited research explores integration with the mobile money services widely used in Tanzania. This system addresses this gap by combining all these components into a single, scalable, cloud-based platform designed for multi-location marketplace deployments.

## CHAPTER THREE: DATA ANALYSIS AND PRESENTATION

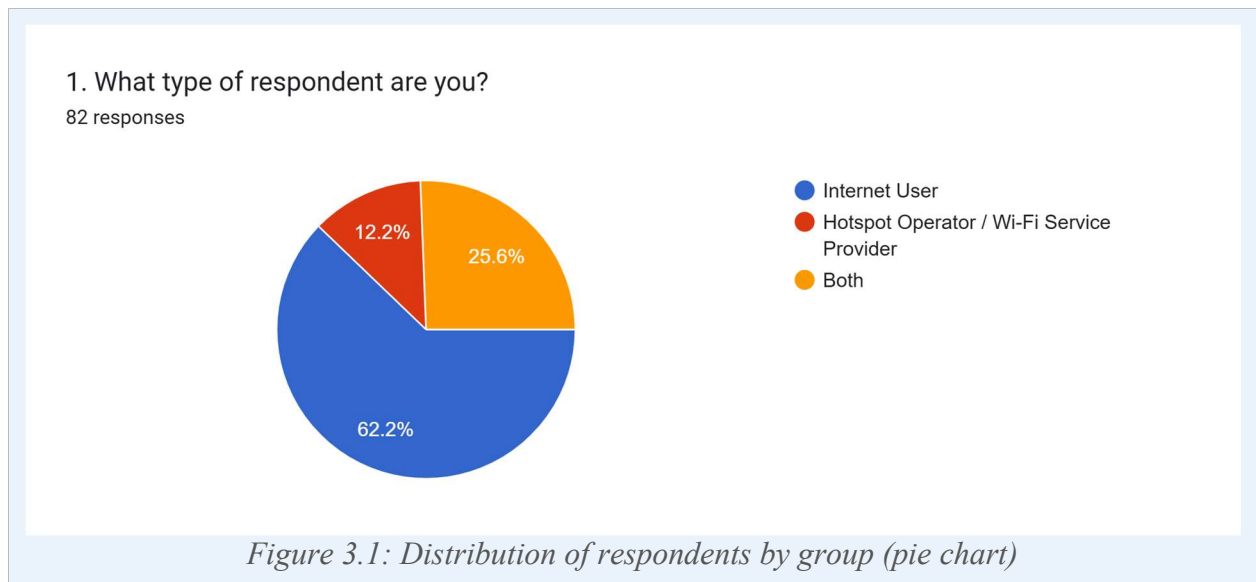
### 3.1 Introduction

This chapter presents and analyzes the data collected to validate the need for the system. Data were collected using a structured closed-ended questionnaire administered online via Google Forms to respondents in Karume Market. A total of 82 responses were received. The analysis is organized so that each specific objective is supported by its own set of questions and findings.

### 3.2 Respondent Profile

The 82 respondents were classified as Internet users, hotspot operators, or both. Table 3.1 summarizes the distribution.

| Respondent Group         | Number | Percentage (%) |
|--------------------------|--------|----------------|
| Internet Users only      | 51     | 62.2%          |
| Hotspot Operators only   | 10     | 12.2%          |
| Both (User and Operator) | 21     | 25.6%          |
| Total                    | 82     | 100%           |



For analysis purposes, the 21 respondents who identified as 'Both' are included in both user-side and operator-side calculations, giving 72 user-side responses and 31 operator-side responses.

### 3.3 Findings for Objective 1: Centralized Cloud Management

Objective 1 requires a cloud-based platform for centralized, remote management of multiple hotspots with revenue and usage reporting. Operators were asked how they manage their routers, whether they use a centralized system, and whether centralized management would improve their operations.

| <b>Management Method</b>           | <b>Frequency</b> | <b>Percentage (%)</b> |
|------------------------------------|------------------|-----------------------|
| Manual configuration               | 7                | 36.8%                 |
| Router management software (local) | 12               | 63.2%                 |
| Centralized system                 | 0                | 0%                    |
| Total                              | 19               | 100%                  |

Table 3.2: How operators currently manage their routers (Q9)

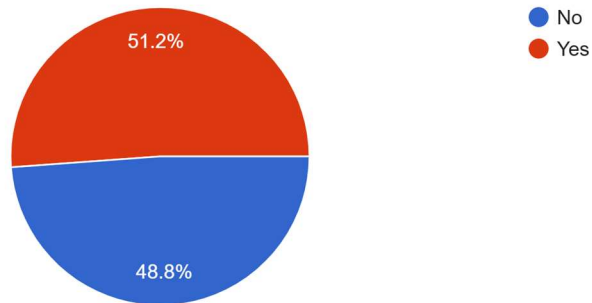
| <b>Use Centralized System</b> | <b>Frequency</b> | <b>Percentage (%)</b> |
|-------------------------------|------------------|-----------------------|
| Yes                           | 9                | 47.4%                 |
| No                            | 10               | 52.6%                 |
| Total                         | 19               | 100%                  |

Table 3.3: Use of a centralized management system (Q10)

52.6% of operators confirmed they do not use a centralized system, while 94.7% agreed that centralized hotspot management would improve their operations (Q12). No operator reported using a fully centralized platform. These findings confirm the need for a cloud-based centralized management system with remote access and reporting as proposed in Objective 1.

10. Do you use a centralized system to manage multiple routers?

43 responses



12. Do you think centralized hotspot management would improve your operations?

43 responses

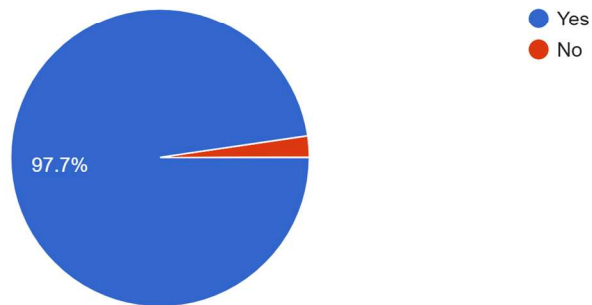


Figure 3.2: Operator views on centralized management (bar chart, Q10 and Q12)

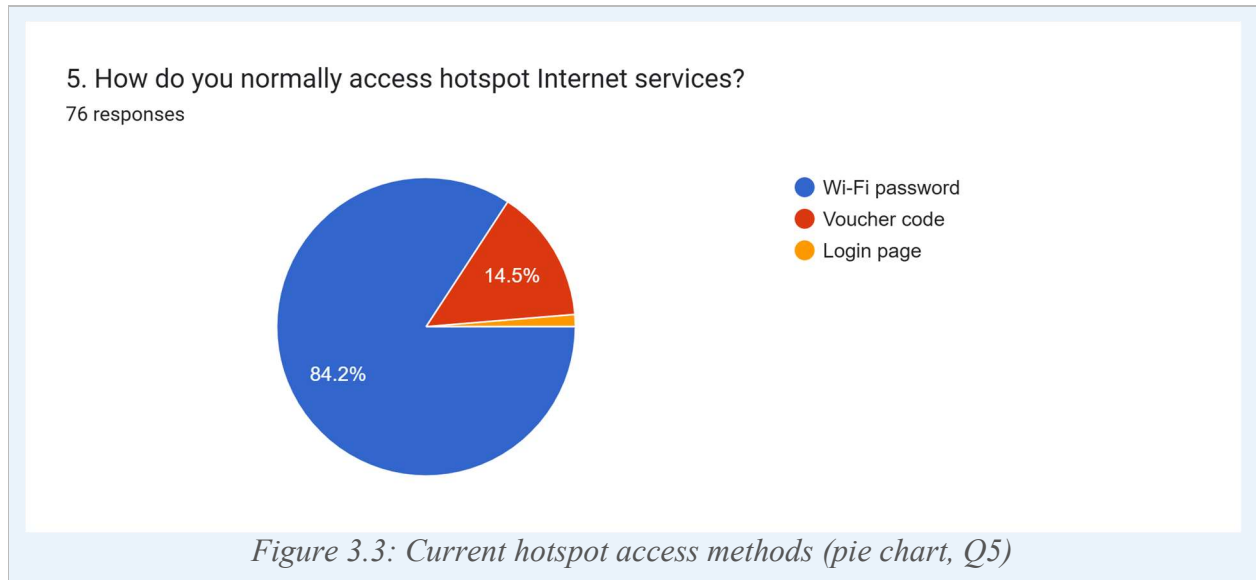
### 3.4 Findings for Objective 2: Captive Portal and Billing

Objective 2 concerns authenticated access through a captive portal with automated billing. Users were asked how they currently access the hotspot, revealing the prevalence of insecure shared-password methods.

| Access Method               | Frequency | Percentage (%) |
|-----------------------------|-----------|----------------|
| Wi-Fi password (shared)     | 60        | 83.3%          |
| Voucher code                | 11        | 15.3%          |
| Login page (captive portal) | 1         | 1.4%           |
| Total                       | 72        | 100%           |

Table 3.4: How users currently access hotspot services (Q5)

83.3% of user-side respondents currently rely on shared Wi-Fi passwords, which is the least secure access method. Only 1 respondent (1.4%) uses a proper captive portal login page. This demonstrates a critical gap in authentication and supports the development of a captive portal with automated billing as proposed in Objective 2.



### 3.5 Findings for Objective 3: Mobile Money Payment Integration

Objective 3 concerns integrating mobile money payment for self-service Internet access. Users were asked which payment method they currently use and whether they experience payment difficulties.

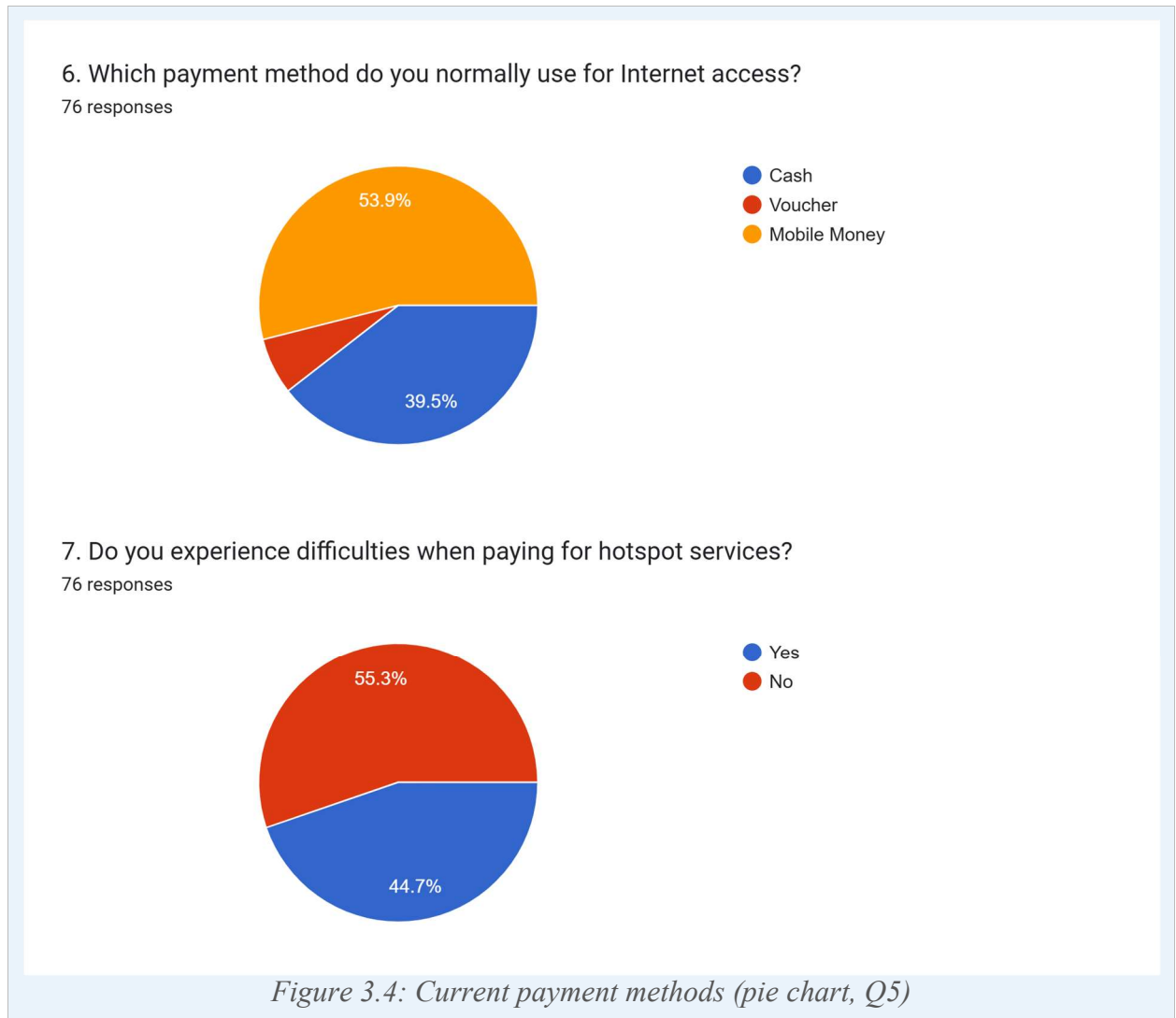
| Current Payment Method | Frequency | Percentage (%) |
|------------------------|-----------|----------------|
| Cash                   | 28        | 38.9%          |
| Mobile Money           | 39        | 54.2%          |
| Voucher                | 5         | 6.9%           |
| Total                  | 72        | 100%           |

Table 3.5: Current payment methods used by users (Q5)

| Experienced Payment Difficulties | Frequency | Percentage (%) |
|----------------------------------|-----------|----------------|
| Yes                              | 32        | 44.4%          |
| No                               | 40        | 55.6%          |
| Total                            | 72        | 100%           |

Table 3.6: Whether users experience payment difficulties (Q6)

54.2% of users already use mobile money for payment, indicating high adoption of this method. However, 44.4% still experience difficulties or delays with current payment processes. Furthermore, 78.9% of operators agreed that automated billing systems would improve hotspot service management (Q13), and 96.3% of all respondents agreed that Internet access should be activated automatically after payment (Q18). These results strongly validate the need for automated mobile money payment integration as proposed in Objective 3.



### 3.6 Findings for Objective 4: Router and Network Monitoring with Alerts

Objective 4 requires real-time monitoring of router availability, connected users, and bandwidth, with automatic alerts on failure. Operators were asked about their current monitoring practices and whether they receive failure alerts.

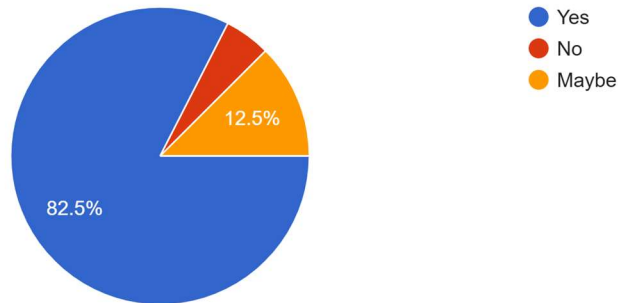
| <b>Number of Routers Managed</b> | <b>Frequency</b> | <b>Percentage (%)</b> |
|----------------------------------|------------------|-----------------------|
| 1 router                         | 12               | 63.2%                 |
| 1–5 routers                      | 4                | 21.1%                 |
| More than 5 routers              | 3                | 15.8%                 |
| Total                            | 19               | 100%                  |

Table 3.7: Number of routers managed per operator (Q8)

36.8% of operators rely entirely on manual configuration, and none reported having real-time centralized monitoring. 82.5% of all respondents agreed that a cloud-based system would improve Internet service reliability (Q16), and 86.3% agreed that automatic router failure notifications would improve reliability (Q17). These findings confirm the need for a router and network monitoring module with alerts, as proposed in Objective 4.

16. Would a cloud-based hotspot management system improve Internet service reliability?

80 responses



17. Would hotspot systems automatically notify administrators when routers go offline improve reliability?

80 responses

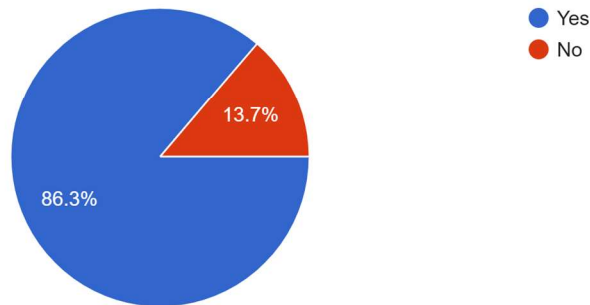


Figure 3.5: Operator agreement on monitoring and alerts (bar chart, Q16 and Q17)

### 3.7 Findings for Objective 5: Congestion Detection

Objective 5 concerns a rule-based module for detecting and predicting network congestion. The open-ended responses (Q19) provide strong qualitative evidence for this need. Multiple respondents independently reported congestion as a key challenge.

| Congestion-Related Comment (Q19 — selected responses)                          | Respondent Type |
|--------------------------------------------------------------------------------|-----------------|
| 'Congested network'                                                            | Both            |
| 'Slow of network'                                                              | Both            |
| 'The higher the internet users the lower the internet speed. This is tiresome' | Internet User   |
| 'When hotspot connected to many users, the internet service running slowly'    | Internet User   |

| <b>Congestion-Related Comment (Q19 — selected responses)</b>                      | <b>Respondent Type</b> |
|-----------------------------------------------------------------------------------|------------------------|
| 'Internet speed should be improved when number of users increase'                 | Internet User          |
| 'Some challenges with hotspot internet services are slow speed during peak hours' | Both                   |
| 'Many people are using the internet, so the network is slow'                      | Both                   |
| 'Challenge is slow internet speed'                                                | Hotspot Operator       |

Table 3.8: Selected qualitative responses on congestion and network speed (Q19)

These consistent reports of slow speeds during peak usage hours provide qualitative validation for the congestion-detection module. The data shows that network congestion is a real, widely-experienced problem that affects both users and operators, supporting the inclusion of a simple rule-based congestion prediction mechanism as proposed in Objective

### 3.8 General Validation

Beyond the objective-specific findings, the general system improvement questions provide overall validation. 92.5% of respondents agreed that hotspot services in marketplaces need better management systems (Q15), and 73.7% of operators expressed willingness to adopt a cloud-based management system (Q14). These results confirm broad support for developing Proposed system and provide a strong go ahead for the project.

| <b>Question</b>                                       | <b>Yes / Agree</b> | <b>Maybe</b> | <b>No</b> |
|-------------------------------------------------------|--------------------|--------------|-----------|
| Q12: Would centralized management improve operations? | 94.7%              | —            | 5.3%      |
| Q13: Would automated billing improve management?      | 78.9%              | 15.8%        | 5.3%      |
| Q14: Willing to use a cloud-based system?             | 73.7%              | 15.8%        | 10.5%     |
| Q15: Hotspot services need better management systems? | 92.5%              | 3.8%         | 3.8%      |
| Q16: Cloud system would improve reliability?          | 82.5%              | 12.5%        | 5.0%      |
| Q17: Automatic alerts would improve reliability?      | 86.3%              | —            | 13.8%     |
| Q18: Auto-activation after payment preferred?         | 96.3%              | —            | 3.8%      |

Table 3.9: Summary of general validation responses

### 3.9 Summary of Findings

The analysis demonstrates that current hotspot management practices in Karume Market are largely manual, insecure, and difficult to scale. All five specific objectives are supported by real data:

| <b>Objective</b>                | <b>Key Finding</b>                                                  | <b>Status</b> |
|---------------------------------|---------------------------------------------------------------------|---------------|
| Obj 1: Centralized management   | 52.6% lack a centralized system; 94.7% want it                      | Confirmed ✓   |
| Obj 2: Captive portal & billing | 83.3% rely on insecure shared passwords                             | Confirmed ✓   |
| Obj 3: Mobile money payment     | 44.4% experience payment difficulties; 96.3% want auto-activation   | Confirmed ✓   |
| Obj 4: Monitoring & alerts      | 86.3% want automatic failure alerts; none have real-time monitoring | Confirmed ✓   |
| Obj 5: Congestion detection     | Multiple respondents cite peak-hour congestion as a key challenge   | Confirmed ✓   |

## CHAPTER FOUR: SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

### 4.1 Introduction

This chapter specifies the software requirements for the proposed system. It defines the purpose, scope, system users, functional requirements, non-functional requirements, and system constraints that guide the design and implementation. The requirements were derived from the problem statement, the study objectives, the data collected, and analysis of existing hotspot practices.

### 4.2 Purpose and Scope

The purpose of proposed system is to provide a centralized, cloud-based platform through which hotspot operators can manage multiple Wi-Fi routers, authenticate users, automate billing through mobile money, monitor network performance, and receive alerts on router failures. The scope covers: the captive portal used by Internet users; the administrative dashboard used by operators; the cloud backend services; and integration with MikroTik routers, FreeRADIUS, mobile money, intelligently detect and predict network congestion to support fair bandwidth management and SMS for one-time passwords.

### 4.3 System Users

| User                     | Description                                                                        | Main Interactions                                                                              |
|--------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Internet User            | A trader, customer, or visitor who connects to the hotspot to access the Internet. | Connects to Wi-Fi, verifies phone via OTP, selects a plan, pays via mobile money, and browses. |
| Administrator / Operator | The hotspot provider who owns and manages the service.                             | Manages plans, customers, devices, payments, sessions, routers; views reports and alerts.      |
| System (Automated)       | Scheduled background processes within the system.                                  | Expires sessions, checks router heartbeats, marks routers offline, raises alerts.              |

### 4.4 Functional Requirements

#### 4.4.1 Captive Portal and Authentication

- I. The system shall detect a connecting device by its MAC address and redirect it to the captive portal.
- II. The system shall verify a user's phone number using a one-time password (OTP) sent by SMS.
- III. The system shall register a device and link it to a customer account.
- IV. The system shall grant Internet access only after successful authentication and payment.

#### **4.4.2 Plans and Billing**

- I. The system shall allow the administrator to create, update, activate, and deactivate Internet access plans with a price, duration, and bandwidth limits.
- II. The system shall display available plans to the user on the captive portal.
- III. The system shall record each payment with its amount, provider, reference, and status.
- IV. The system shall automatically activate the selected plan upon successful payment confirmation.

#### **4.4.3 Mobile Money Payment**

- I. The system shall initiate a mobile money payment for a selected plan.
- II. The system shall verify payment confirmation from the payment provider before granting access.
- III. The system shall support Tanzanian mobile money services such as M-Pesa and Airtel Money through a payment provider interface.

#### **4.4.4 Session Management**

- I. The system shall create a time-bound access session upon plan activation.
- II. The system shall automatically expire sessions when their time elapses.
- III. The system shall allow the administrator to view and manually terminate active sessions.

#### **4.4.5 Router and Network Monitoring**

- I. The system shall receive periodic heartbeats from each router and record its status, connected clients, and resource usage.
- II. The system shall mark a router as degraded or offline when heartbeats are missed for defined intervals.
- III. The system shall track bandwidth usage and the number of connected users per router.
- IV. The system shall notify the administrator when a router becomes unavailable.

#### **4.4.6 Congestion Detection**

- I. The system shall analyze recorded usage data to identify periods of high network load on a router.
- II. The system shall flag potential congestion when the number of connected users or bandwidth usage exceeds a defined threshold, and indicate likely busy periods to the administrator.

#### **4.4.7 Administration and Reporting**

- I. The system shall authenticate administrators using secure login credentials with JWT.

- II. The system shall provide a dashboard showing key indicators such as total customers, devices, routers, active sessions, and revenue.
- III. The system shall allow management of customers, devices, plans, payments, sessions, routers, and sites.
- IV. The system shall generate reports on revenue and network usage.
- V. The system shall allow administrators to manage the system remotely over the Internet.

#### 4.5 Non-Functional Requirements

| Requirement     | Description                                                                                                                                                           |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Security        | Administrator access protected by JWT authentication and encrypted credentials; user authentication uses OTP rather than shared passwords to prevent revenue leakage. |
| Reliability     | The system detects router failures and alerts administrators to minimize downtime.                                                                                    |
| Scalability     | The cloud architecture supports adding more routers and locations without redesign.                                                                                   |
| Usability       | The captive portal is usable on a phone browser without installing software; the admin dashboard is clear and easy to navigate.                                       |
| Performance     | Session activation and payment confirmation are processed promptly so access is granted without noticeable delay.                                                     |
| Availability    | The cloud server is accessible over the Internet so operators can manage the system remotely at any time.                                                             |
| Maintainability | The system uses a modular structure so that components such as payment or SMS providers can be replaced or extended.                                                  |

#### 4.6 System Constraints

- **Router compatibility.** The system requires MikroTik routers for enforcement.
- **Connectivity.** The system requires Internet connectivity between the routers and the cloud server.
- **External services.** Mobile money and SMS functionality depend on the availability of the respective provider gateways.
- **Testing environment.** Evaluation is carried out in a controlled or simulated environment rather than full production infrastructure.

## CHAPTER FIVE: SYSTEM DESIGN

### 5.1 Introduction

This chapter presents the design of system based on the requirements in the previous chapter. It covers the system architecture, data flow, use cases, component interactions, database design, and user interface design. The design separates network management in the cloud from network enforcement at the routers, in line with the AAA framework.

### 5.2 System Architecture

The proposed system adopts a three-tier, cloud-centered architecture:

- **Enforcement tier.** MikroTik hotspot routers that connect users, redirect them to the captive portal, and enforce access policies and bandwidth limits based on instructions from the cloud.
- **Cloud application tier.** A Spring Boot backend exposing APIs and a React frontend providing the captive portal and the administrative dashboard. This tier handles authentication, billing, session management, monitoring, congestion detection, and reporting.
- **Data tier.** A PostgreSQL database that stores customers, devices, plans, payments, sessions, routers, RADIUS accounts, and monitoring data.
- **External integrations.** FreeRADIUS for authentication and accounting; a mobile money payment gateway for payment processing; and an SMS gateway for OTP delivery.

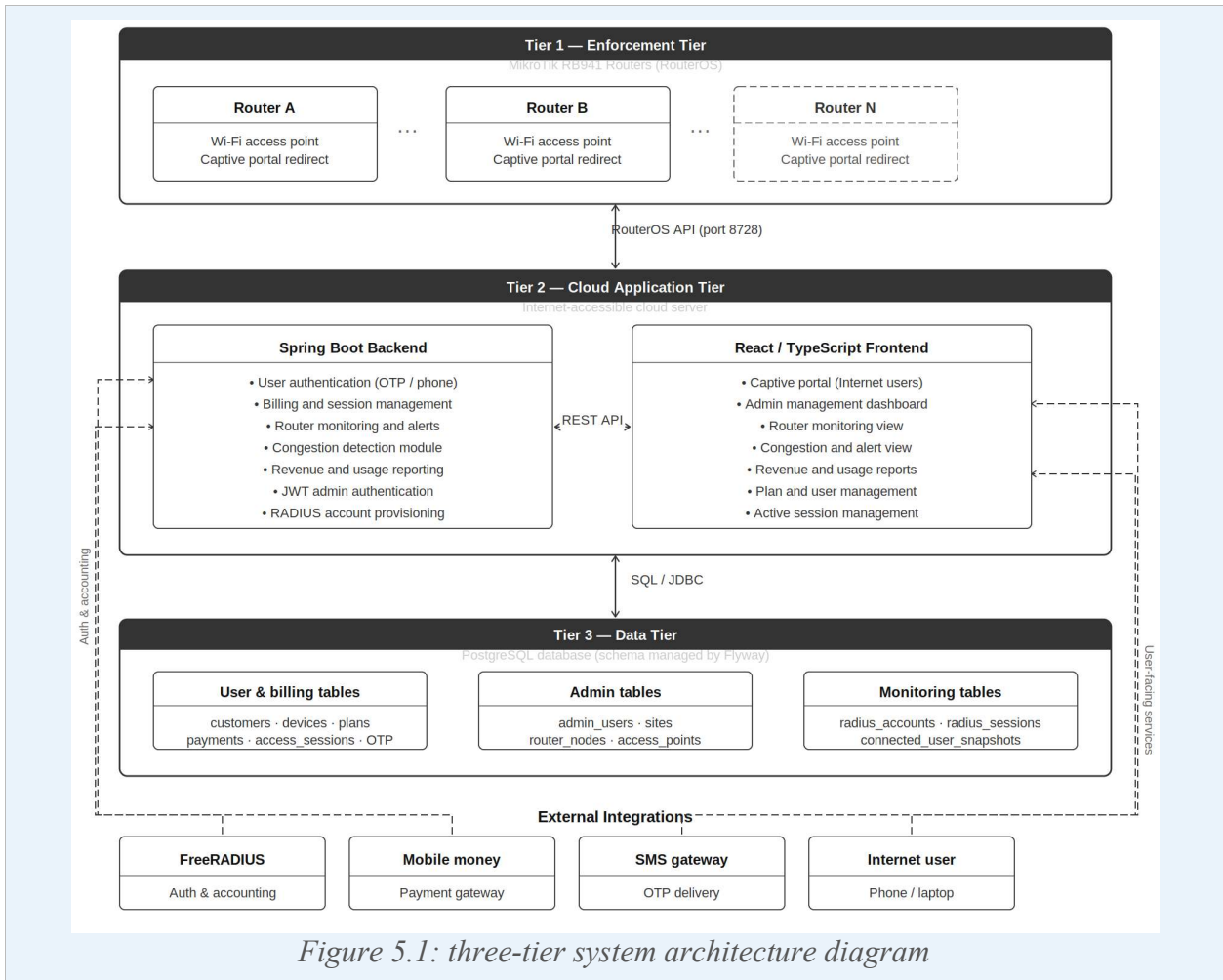
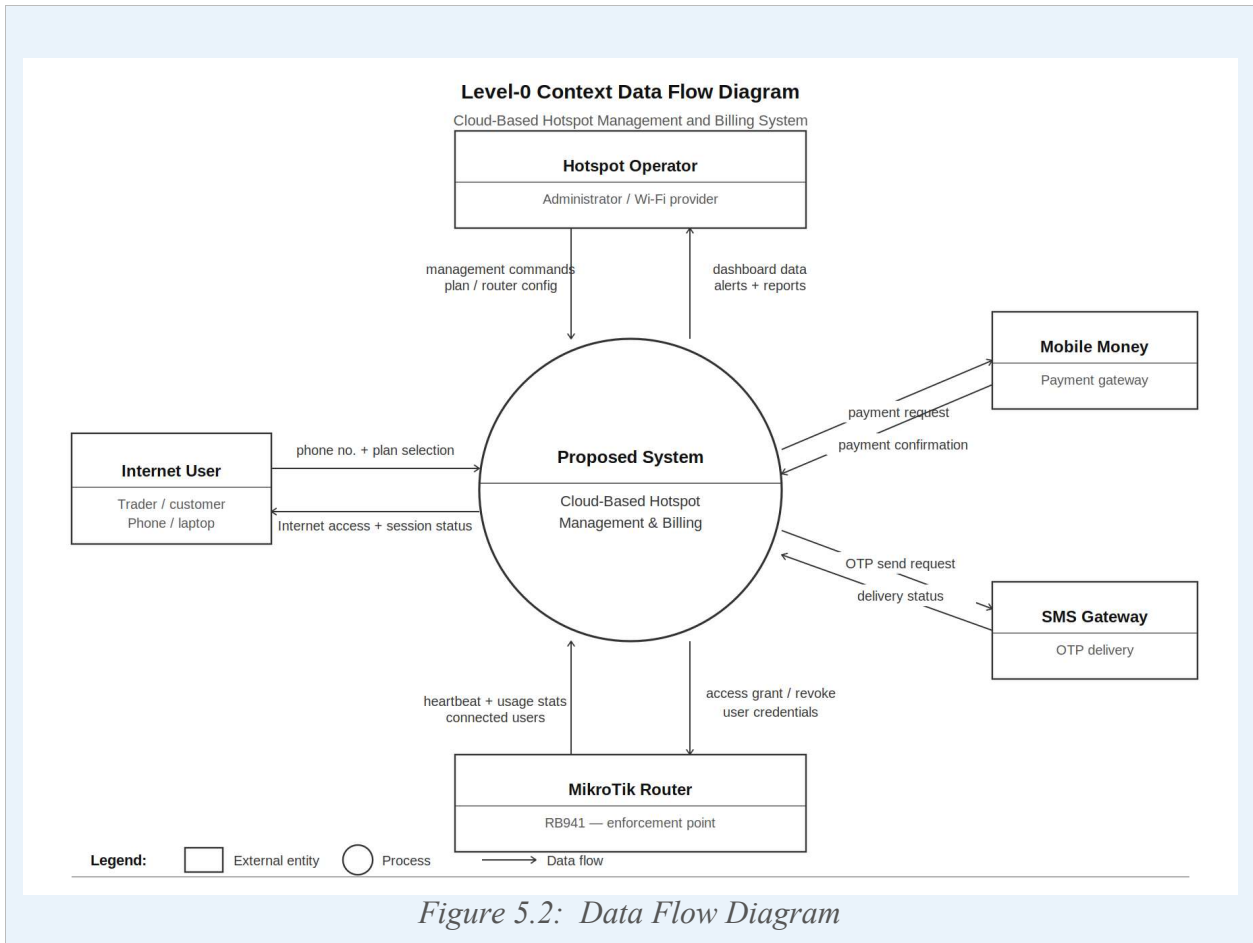


Figure 5.1: three-tier system architecture diagram

### 5.3 Data Flow Diagram

The data flow describes how information moves through the system. A user connecting to a router is redirected to the captive portal. The portal sends the user's phone number and selected plan to the cloud server, which authenticates the user via OTP, initiates and verifies mobile money payment, activates the plan, provisions RADIUS credentials, and instructs the router to grant access. Session usage and router heartbeat data flow back to the cloud server and are stored in the database for monitoring and reporting.

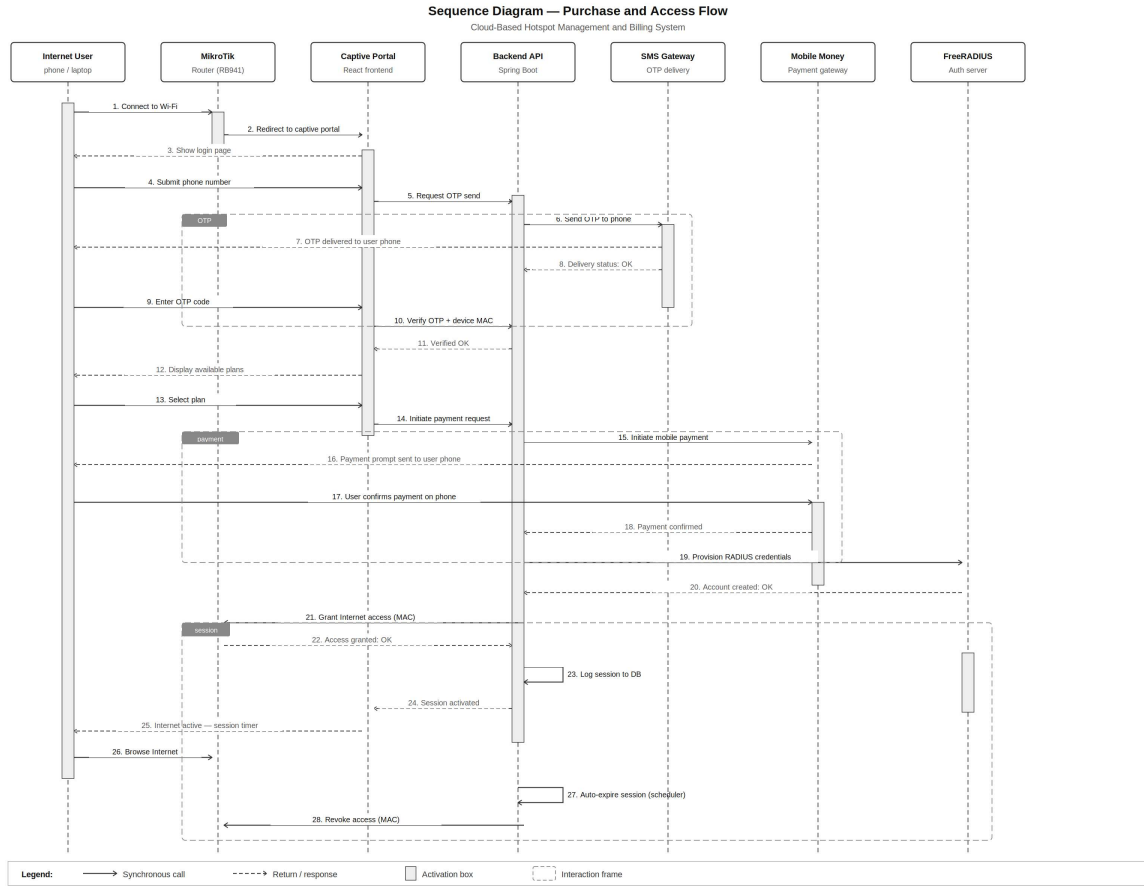


## 5.4 Use Case Diagram

The system has two primary human actors (Internet User and Administrator) and external service actors (mobile money gateway and SMS gateway). The main use cases are summarized in Table 5.1.

| Actor                | Use Cases                                                                                                                                                                                |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Internet User        | Connect to hotspot; verify phone via OTP; select plan; make payment; receive Internet access; view session status and countdown.                                                         |
| Administrator        | Log in; manage plans; manage customers and devices; view payments; manage sessions; manage routers and sites; view monitoring dashboard; receive alerts; view revenue and usage reports. |
| Mobile Money Gateway | Process payment; confirm payment status to the backend.                                                                                                                                  |
| SMS Gateway          | Deliver one-time password to the user's phone.                                                                                                                                           |



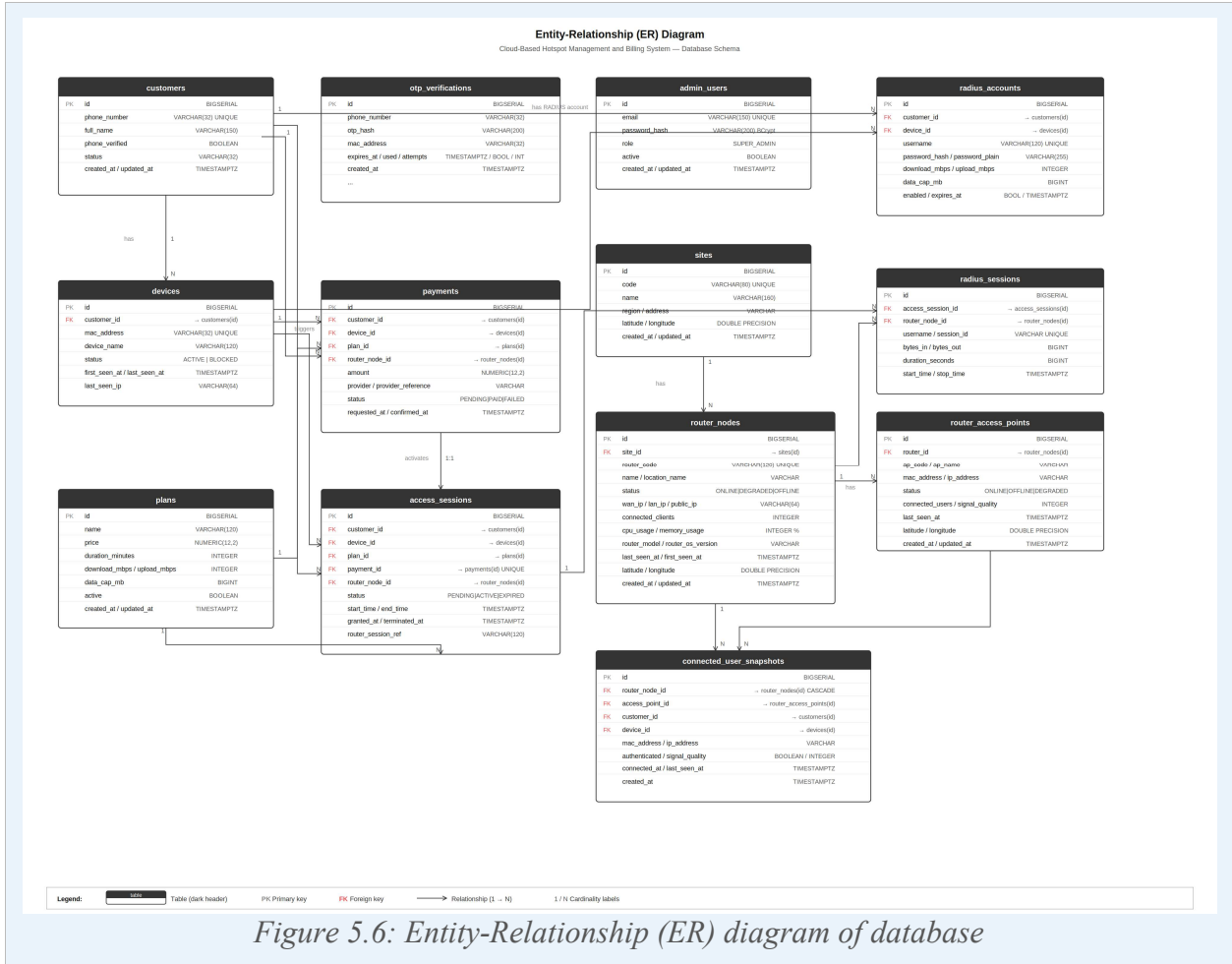


## 5.6 Database Design

The database is implemented in PostgreSQL and consists of interrelated tables for all system entities. Table 5.2 summarizes the core tables, their purpose, and their relationships.

| Table           | Purpose                                                                   | Key Relationships                                    |
|-----------------|---------------------------------------------------------------------------|------------------------------------------------------|
| customers       | Stores customer identity, phone number, and OTP verification.             | One customer → many devices, payments, and sessions. |
| devices         | Stores user devices identified by MAC address.                            | Each device belongs to one customer.                 |
| plans           | Stores Internet access plans, prices, duration, and bandwidth limits.     | Referenced by payments and access sessions.          |
| payments        | Records payment transactions, amounts, providers, references, and status. | Links a customer, device, and plan.                  |
| access_sessions | Time-bound Internet access sessions granted after payment.                | Links a payment, customer, device, and plan.         |

| <b>Table</b>             | <b>Purpose</b>                                                         | <b>Key Relationships</b>                   |
|--------------------------|------------------------------------------------------------------------|--------------------------------------------|
| otp_verifications        | Stores hashed OTP codes for phone verification.                        | Linked by phone number and device MAC.     |
| admin_users              | Stores administrator credentials and roles.                            | Used for dashboard authentication via JWT. |
| router_nodes             | Stores router identity, status (ONLINE/DEGRADED/OFFLINE), and metrics. | Belongs to a site; has access points.      |
| router_access_points     | Stores Wi-Fi access points of a router.                                | Each belongs to one router.                |
| sites                    | Groups routers by location or deployment site.                         | One site → many routers.                   |
| radius_accounts          | FreeRADIUS credentials with bandwidth limits.                          | Linked to a customer and device.           |
| radius_sessions          | RADIUS accounting records (bytes in/out, session duration).            | Linked to an access session and router.    |
| connected_user_snapshots | Periodic snapshots of connected users per router.                      | Linked to a router and access point.       |



## 5.7 Congestion Detection Module Design

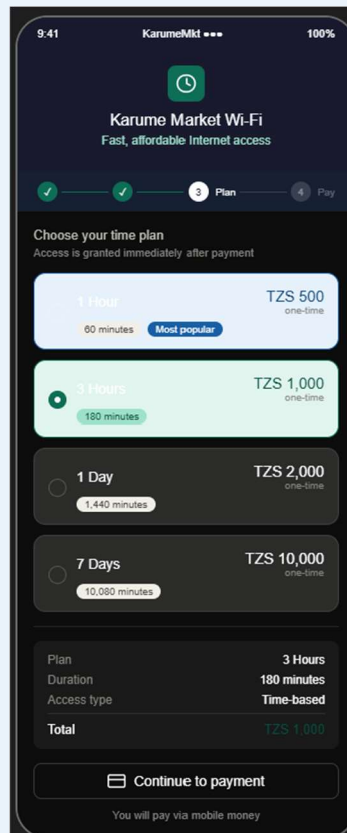
The congestion detection module uses a rule-based statistical approach to identify and predict network overload. The module operates as follows:

- vi. At regular intervals, the system reads `connected_user_snapshots` and `radius_sessions` to obtain the current load (number of connected users, bytes in/out) for each router.
- vii. For each router, the system maintains a moving average of connected users and bandwidth consumption over a configurable rolling window (for example, the past 24 hours in hourly slots).
- viii. If the current load exceeds a defined threshold (for example, 80% of the router's rated capacity), the system flags the router as congested and records the event.
- ix. From the historical averages, the system identifies time slots that consistently exceed the threshold. These are presented to the administrator on the dashboard as predicted busy periods.

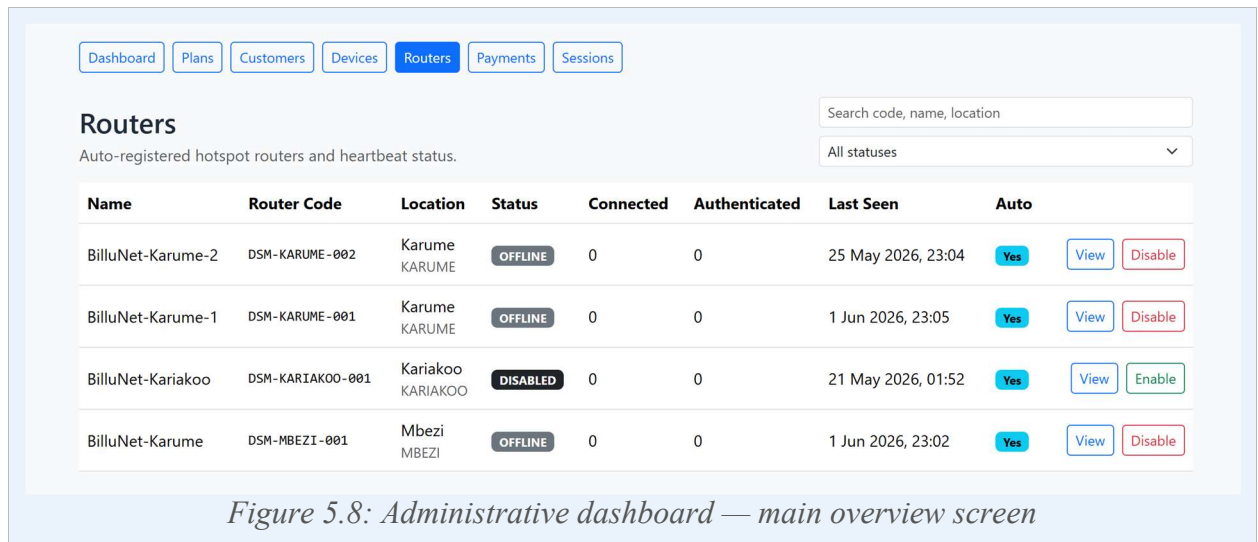
This approach can operate effectively from the usage data the system already collects. The administrator can use the predictions to plan bandwidth allocation or schedule maintenance outside peak periods.

## 5.8 User Interface Design

The system provides two interfaces. The captive portal is the user-facing interface, designed to be simple and usable on a phone browser without installing any software. It guides the user through phone verification (OTP entry), plan selection with prices and durations displayed clearly, mobile money payment initiation, and an active-session view showing remaining time. The administrative dashboard is the operator-facing interface, presenting key performance indicators and providing management screens for all system entities: customers, devices, plans, payments, sessions, routers, access points, connected users, and reports.



*Figure 5.7: Captive portal interface — plan selection page*



## 5.9 Chapter Summary

This chapter presented the design of system, including its three-tier cloud architecture, data flow diagrams, use cases, database structure, congestion detection module design, and user interface layout. The design fulfils all functional and non-functional requirements specified in the previous chapter and provides the complete blueprint for implementing the system.

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